

Control work 8, and work on the mistakes

The probability block in one paper, then the two rules drawn as a single decision tree.

LESSON 97–98

2 × 40 MIN

ALGEBRA 10, NAZORAT ISHI 8

IGX 24 REVIEW

LESSON CLOCK

3

42'

12'

25'

8'

Instructions	3 min
The paper	42 min
Answers	12 min
Diagnosis and rewrite	25 min
The decision tree	8 min

LEARNING OBJECTIVES

- Compute probabilities from a sample space, a table and a tree under time.
- Choose between the addition rule, the multiplication rule and the complement.
- Compute a conditional probability with the right denominator.
- Name each lost mark and rewrite the solution in full.

TEXTBOOK

National	pp. 335–338
Cambridge	Core & Extended, pp. 643–646
Workbook	Control paper A8

01

Explanation

The paper — 30 marks, 42 minutes

Q	TASK	MARKS	FROM
1	Two dice: find $P(\text{total } 8)$, $P(\text{a double})$, $P(\text{total } 8 \text{ or a double})$	6	L91–96
2	A coin thrown 5 times: find $P(\text{at least one head})$ and $P(\text{exactly one head})$	5	L94–96
3	A bag of 5 red and 4 green, two drawn without replacement: the tree, then $P(\text{same colour})$	6	L91–96
4	Of 50 pupils, 32 study physics, 21 chemistry, 15 both. Find $P(\text{neither})$ and $P(\text{physics} \mid \text{chemistry})$	7	L94–96
5	A point in a square of side 8 lands inside the inscribed circle: find P	3	L94–96
6	State whether two given events are exclusive, independent, or neither, with a reason	3	L91–93

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for subtracting the overlap, Q2 one for using the complement, Q3 one for changing the second fork, Q4 one for the denominator 21 rather than 50. Four of the thirty marks are for a single decision each.

The decision tree for the whole block

THE QUESTION SAYS	USE	FORMULA
“or”, “either”	the addition rule	$P(A) + P(B) - P(A \cap B)$
“and”, “both”, “then”	the multiplication rule	$P(A) \cdot P(B \mid A)$
“at least one”	the complement	$1 - P(\text{none})$
“given that”, “of those who”	conditional	$\frac{P(A \cap B)}{P(B)}$
“without replacement”	a tree with changing forks	second row differs
“at random inside”	geometric probability	$\frac{\text{area}}{\text{area}}$

THE FOUR SLIPS THAT ACCOUNT FOR MOST LOST MARKS

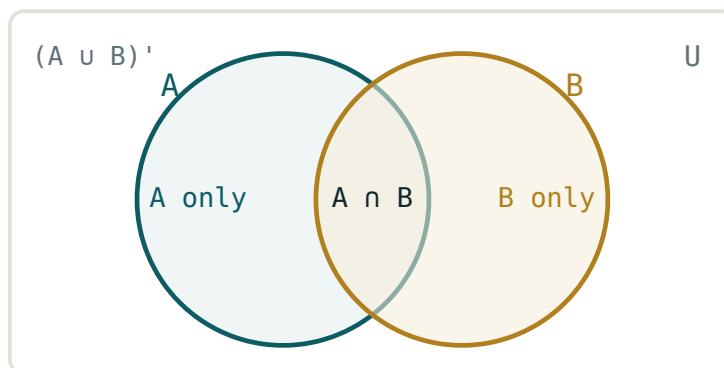
- adding without subtracting the overlap;
- multiplying dependent events as though they were independent;
- using $|U|$ as the denominator of a conditional probability;
- forgetting that “without replacement” changes the second fork.

Looking forward

Lessons 99–102 are the Cambridge revision block: sequences, arithmetic and geometric progressions, infinite geometric series and the binomial expansion. None of it is new to the national syllabus of Grade 9, but the notation is Cambridge's and the questions are harder.

ONE HABIT TO CARRY FORWARD

Probability rewarded writing the sample space before anything else. Series reward writing a and d (or r) before anything else. In both, one line of setting-up prevents three lines of error.



$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The picture behind Q1 and Q4 alike.

02 Worked examples

EXAMPLE 1 Model answer, Q1: two dice, $P(\text{total 8 or a double})$.

① Total 8: $(2,6),(3,5),(4,4),(5,3),(6,2) — 5$.

② Doubles: 6.

③ Both: $(4,4) — 1$.
The overlap.

④
$$P = \frac{5}{36} + \frac{6}{36} - \frac{1}{36} = \frac{10}{36} = \frac{5}{18}$$

ANSWER

$$\frac{5}{18} \approx 0.278$$

EXAMPLE 2 Model answer, Q3: 5 red and 4 green, two drawn without replacement, $P(\text{same colour})$.

①
$$P(RR) = \frac{5}{9} \times \frac{4}{8} = \frac{5}{18}$$

②
$$P(GG) = \frac{4}{9} \times \frac{3}{8} = \frac{1}{6}$$

③
$$P = \frac{5}{18} + \frac{3}{18} = \frac{8}{18} = \frac{4}{9}$$

Add down the ends.

ANSWER

$$\frac{4}{9} \approx 0.444$$

EXAMPLE 3 Model answer, Q4: 50 pupils, 32 physics, 21 chemistry, 15 both.

$$① \quad |P \cup C| = 32 + 21 - 15 = 38$$

$$② \quad P(\text{neither}) = \frac{12}{50} = 0.24$$

$$③ \quad P(\text{physics} \mid \text{chemistry}) = \frac{15}{21} = \frac{5}{7}$$

Denominator 21.

ANSWER

0.24 and $\frac{5}{7}$

03 Interactive model

Work Q4 twice — once with denominator 50, once with 21 — and let the class say which question each answers.

Ten questions on the probability block

Two dice give how many outcomes?

12

36

21

6

$P(A')$ is:

$1 - P(A)$

$P(A)$

0

$1 + P(A)$

The addition rule subtracts:

nothing

the overlap

the smaller probability

1

Independent events multiply because:

they overlap

one does not affect the other

they are exclusive

of the tree

“At least one” suggests:

a long list

the complement

geometry

conditioning

$P(B | A)$ divides by:

$P(B)$

$P(A)$

1

$|U|$

“Without replacement” changes:

nothing

the second fork

the sample space size only

the first fork

Geometric probability divides:

counts

measures

frequencies

nothing

Expected sixes in 600 throws:

60

100

120

600

Exclusive events with positive probability are:

independent

not independent

certain

complementary

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Sample space	Elementar hodisalar fazosi	Пространство исходов
Addition rule	Qo'shish qoidasi	Правило сложения
Multiplication rule	Ko'paytirish qoidasi	Правило умножения
Complement	To'ldiruvchi	Дополнение
Conditional probability	Shartli ehtimollik	Условная вероятность
Without replacement	Qaytarmasdan	Без возвращения
Target	Maqsad	Цель

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The overlap in Q1 is:

(4,4)

(2,6)

none

all doubles

Q3 needs a tree because:

there are two colours

the second draw depends on the first

it is longer

no reason

The denominator in Q4's conditional part is:

50

32

21

15

Lessons 99–102 revise:

trigonometry

progressions and the binomial expansion

geometry

logarithms

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. P two dice total 8

ANSWER $\frac{5}{36}$

2. P two dice show a double

ANSWER $\frac{1}{6}$

3. P at least one head in five throws

ANSWER $\frac{31}{32}$

4. P exactly one head in five throws

ANSWER $\frac{5}{32}$

5. A point in a square of side 8 hits the inscribed circle

ANSWER $\frac{\pi}{4} \approx 0.785$

6. $P(A \cup B)$ when $P(A) = 0.4$, $P(B) = 0.3$ and exclusive

ANSWER 0.7

7. Are “even” and “multiple of 3” on a die exclusive?

ANSWER No — 6 is both

Medium · the standard question

7 PROBLEMS

1. P two dice: total 8 or a double

ANSWER $\frac{5}{18}$

2. 5 red, 4 green, two drawn without replacement: $P(RR)$

ANSWER $\frac{5}{18}$

3. Same: P same colour

ANSWER $\frac{4}{9}$

4. Same: P different colours

ANSWER $\frac{5}{9}$

5. 50 pupils, 32 physics, 21 chemistry, 15 both: P neither

ANSWER 0.24

6. Same: $P(\text{physics} \mid \text{chemistry})$

ANSWER $\frac{5}{7}$

7. Same: $P(\text{chemistry} \mid \text{physics})$

ANSWER $\frac{15}{32}$

Hard · stretch and reason

7 PROBLEMS

1. P at least two heads in four throws

ANSWER $\frac{11}{16}$

2. Three cards from 52 without replacement: P all hearts

ANSWER $\frac{11}{850}$

3. Two dice: $P(\text{total } 8 \mid \text{first is even})$

ANSWER $\frac{1}{6}$

4. A bag of 6 white and 4 black: P the second is white

ANSWER $\frac{3}{5}$

5. A point in a circle of radius 5: P it lies within 3 of the centre

ANSWER 0.36

6. If $P(A) = 0.7$, $P(B) = 0.4$, $P(A \cup B) = 0.82$, are they independent?

ANSWER Yes — $P(A \cap B) = 0.28 = 0.7 \times 0.4$

7. A die is thrown until a six appears: P it takes at most three throws

ANSWER $\frac{91}{216} \approx 0.421$

07 Homework

Homework — 4 tasks

Bring the decision tree to Lesson 99; the revision block assumes it is done.

1. Rewrite in full every control-work question that lost a mark, naming the slip in the margin.
2. Copy the decision-tree table and add one worked example of your own to each row.

3. A bag holds 7 blue and 3 yellow counters; three are drawn without replacement. Find the probability that all three are blue.
4. Write your target for the revision block in one checkable sentence, and date it.